

Question Paper: Machine Learning

****Machine Learning Question Paper****

****Class 9-12, Difficulty: Hard****

****Multiple Choice Questions (1-10)****

1. What is the primary goal of supervised learning in machine learning?

- A) To identify patterns in unlabeled data
- B) To predict outcomes based on labeled data
- C) To optimize model performance using reinforcement learning
- D) To visualize high-dimensional data

Answer: ****B) To predict outcomes based on labeled data****

2. Which of the following algorithms is commonly used for clustering in machine learning?

- A) Linear Regression
- B) Decision Trees
- C) K-Means
- D) Support Vector Machines

Answer: ****C) K-Means****

3. What is the term for the process of reducing the dimensionality of high-dimensional data?

- A) Feature extraction
- B) Feature selection

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C) Dimensionality reduction

D) Data augmentation

Answer: **C) Dimensionality reduction**

4. Which type of machine learning model is trained using both labeled and unlabeled data?

A) Supervised learning

B) Unsupervised learning

C) Semi-supervised learning

D) Reinforcement learning

Answer: **C) Semi-supervised learning**

5. What is the purpose of regularization in machine learning models?

A) To increase model complexity

B) To reduce overfitting

C) To improve model interpretability

D) To increase training time

Answer: **B) To reduce overfitting**

6. Which of the following deep learning architectures is commonly used for natural language processing tasks?

A) Convolutional Neural Networks (CNNs)

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- B) Recurrent Neural Networks (RNNs)
- C) Autoencoders
- D) Generative Adversarial Networks (GANs)

Answer: **B) Recurrent Neural Networks (RNNs)**

7. What is the term for the process of using a pre-trained model as a starting point for a new machine learning task?

- A) Transfer learning
- B) Fine-tuning
- C) Hyperparameter tuning
- D) Model selection

Answer: **A) Transfer learning**

8. Which of the following metrics is commonly used to evaluate the performance of a classification model?

- A) Mean Squared Error (MSE)
- B) Mean Absolute Error (MAE)
- C) Accuracy
- D) F1-score

Answer: **C) Accuracy**

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9. What is the purpose of batch normalization in deep neural networks?

- A) To reduce overfitting
- B) To improve model interpretability
- C) To increase model training speed
- D) To normalize input data

Answer: **D) To normalize input data**

10. Which of the following techniques is used to handle imbalanced datasets in machine learning?

- A) Oversampling the minority class
- B) Undersampling the majority class
- C) SMOTE (Synthetic Minority Over-sampling Technique)
- D) All of the above

Answer: **D) All of the above**

****Short Questions (11-15)****

11. Explain the concept of overfitting in machine learning and provide an example of how it can occur. (4 marks)

12. Describe the difference between supervised and unsupervised learning in machine learning. Provide an example of each. (4 marks)

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13. What is the purpose of feature scaling in machine learning? Explain how it can improve model performance. (3 marks)

14. Compare and contrast the concepts of precision and recall in machine learning. Provide an example of how they are used in a classification problem. (4 marks)

15. Explain the concept of ensemble learning in machine learning. Describe how it can improve model performance and provide an example of a popular ensemble learning algorithm. (5 marks)

****Long/Descriptive Questions (16-18)****

16. Describe the architecture of a Convolutional Neural Network (CNN) and explain how it is used for image classification tasks. Provide an example of a popular CNN architecture and its applications. (10 marks)

17. Explain the concept of reinforcement learning in machine learning. Describe how it is used in robotics and game playing. Provide an example of a popular reinforcement learning algorithm and its applications. (10 marks)

18. Compare and contrast the concepts of support vector machines (SVMs) and random forests in machine learning. Describe how they are used for classification and regression tasks. Provide an example of how they are used in a real-world problem. (10 marks)

****Answer Key****

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MCQs:

1. B) To predict outcomes based on labeled data
2. C) K-Means
3. C) Dimensionality reduction
4. C) Semi-supervised learning
5. B) To reduce overfitting
6. B) Recurrent Neural Networks (RNNs)
7. A) Transfer learning
8. C) Accuracy
9. D) To normalize input data
10. D) All of the above

Short Questions:

11. Overfitting occurs when a model is too complex and fits the training data too closely, resulting in poor performance on new, unseen data. Example: A model that is trained on a small dataset and has a high number of parameters may overfit the training data.
12. Supervised learning involves training a model on labeled data, while unsupervised learning involves training a model on unlabeled data. Example: Supervised learning - image classification, Unsupervised learning - clustering.
13. Feature scaling improves model performance by reducing the effect of dominant features and improving the convergence of optimization algorithms.

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14. Precision is the ratio of true positives to true positives plus false positives, while recall is the ratio of true positives to true positives plus false negatives. Example: In a classification problem, precision and recall can be used to evaluate the performance of a model.

15. Ensemble learning involves combining the predictions of multiple models to improve overall performance. Example: Random forests, boosting.

Long/Descriptive Questions:

16. A CNN consists of convolutional and pooling layers, followed by fully connected layers. It is used for image classification tasks such as object detection and image recognition. Example: AlexNet, VGGNet.

17. Reinforcement learning involves training an agent to take actions in an environment to maximize a reward signal. It is used in robotics and game playing. Example: Q-learning, Deep Q-Networks.

18. SVMs and random forests are both used for classification and regression tasks. SVMs are particularly effective in high-dimensional spaces, while random forests are effective in handling large datasets. Example: SVMs are used in image classification, while random forests are used in recommender systems.